

BEYOND ALL-EQUITY: LEVERED DIVERSIFICATION OVER THE LIFECYCLE

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Abstract

We study whether the case for all-equity lifecycle portfolios survives when households can borrow and invest in global bonds, gold, and managed futures. Without leverage, a multi-asset strategy lowers the probability of financial ruin but not expected utility: it requires a 12.35% savings rate to match an all-equity strategy saving 10%. Levering equities alone adds little utility and raises ruin. Levered to the volatility of unlevered equities, the multi-asset strategy matches the all-equity utility with a 6.35% savings rate, lowers the probability of ruin from 7.0% to 2.3%, and, at the same savings rate, fails to fund the all-equity retirement income less often (3.2% versus 7.0%). The gains come mainly from managed futures, which pay off in equity crises. A strategy of stocks and managed futures alone does better still on every measure, whereas bonds lower utility when they replace stocks.

JEL classifications: C15, D14, G11, G17, G51

Key words: Lifecycle asset allocation, leverage, managed futures, retirement savings, risk parity

Key findings

- Levered to the volatility of equities, a multi-asset portfolio gives a couple the expected utility of an all-equity portfolio with 6.4% of income saved instead of 10%, and cuts its risk of running out of money from 7.0% to 2.3%.
- Managed futures drive the gains; stocks plus managed futures do best. Gold helps since 1970, when its price is free, but not when it was administered. Bonds lower utility when they replace stocks.
- Preserving capital and funding a given income differ: risk parity at 175% runs out of money in 3.8% of lifetimes but fails to pay the all-equity income in 14.2%.

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²The replication code, data, and numerical results are available at <https://github.com/GF-boop/beyond-stocks>. The internet appendix is available as a separate document. The author used OpenAI Codex and Anthropic Claude Code for code development, reproducibility checks, and manuscript editing, and is responsible for the research design, interpretations, code, and final manuscript.

1 Introduction

Anarkulova, Cederburg, and O’Doherty (2025) challenge two central tenets of lifecycle investing. In a lifecycle model with labor income risk, Social Security income, and longevity risk, a US couple that chooses among domestic stocks, international stocks, bonds, and bills optimally holds about one-third domestic stocks and two-thirds international stocks at all ages, with virtually no fixed income. This all-equity strategy dominates balanced portfolios and target-date funds in building wealth, preserving capital, and generating bequests. The conclusion survives leverage. When their couple can borrow at a modest spread above the bill rate, it levers the all-equity portfolio rather than a balanced portfolio of the kind advocated by Asness (1996). Asness (2024) objects that diversified portfolios should be compared with equities at equal risk, after leveraging them, not at equal capital. We test this objection with a broader set of diversifiers.

We add two asset classes that have historically paid off during equity crises, gold and managed futures, alongside currency-hedged global government bonds. These assets have lower expected returns than stocks, so an unlevered household that holds them gives up return. Leverage is therefore the natural companion to diversification: a household that can borrow can hold a more diversified portfolio and scale it up to an equity-like level of risk. This is the portfolio logic behind the leverage-aversion argument of Asness (1996), Frazzini and Pedersen (2014), and Asness, Frazzini, and Pedersen (2012).

We keep the household problem of Anarkulova, Cederburg, and O’Doherty (2025) and implement it at an annual frequency on a public dataset of 16 developed countries from 1927 to 2025. A US couple saves 10% of its labor income from age 25, withdraws 4% of its retirement wealth in real terms from age 65, receives Social Security benefits, and has utility over retirement consumption and bequest. We compare an all-equity strategy, a proportional strategy that holds stocks, global bonds, gold, and managed futures in fixed proportions of 40%, 27%, 17%, and 17%, and a risk parity strategy, financed at the local bill rate plus 0.30%. Following the objection of Asness (2024), our main comparison levers each diversified strategy to the volatility of the unlevered all-equity strategy; we also report exposures from 100% to 200% of wealth.

We report three measures, which answer different questions (Table II). The equivalent savings rate is the savings rate that gives a strategy the expected utility of the all-equity strategy saving 10%. The probability of financial ruin applies the 4% rule to the strategy’s own wealth and does not depend on the savings rate. The path-matched income failure is the probability of not paying the income that the all-equity strategy would have funded on the same simulated path.

Figure 1 summarizes our main results. Without leverage, the proportional strategy lowers the probability of ruin from 7.0% to 5.0% but does not improve expected utility: a couple using it must save 12.35% of its income to match an all-equity couple saving 10%. Levering the all-equity strategy adds little utility and raises its ruin probability at every step. Levered to the volatility of unlevered equities, at 166% exposure, the proportional strategy improves on all three measures. It matches the all-equity utility with a 6.35% savings rate, has a ruin probability of 2.3%, and, saving 10%, fails to fund the path-matched income in 3.2% of lifetimes, compared with 7.0%.

These three numbers should not be combined. The equivalent savings rate is a utility measure: a couple that saves only 6.35% in the proportional strategy obtains the same expected utility, not the same income, and it fails to fund the path-matched income in 13.1% of lifetimes, most often in paths where equities did well and that income is high. A strategy of stocks and managed futures alone, levered to 132%, does better on every measure: it matches the all-equity utility with a 5.22% savings rate and, even at that savings rate, funds the path-matched income more reliably than the all-equity strategy.

These findings are consistent with the portfolio logic of leverage aversion but not with its usual source. In the argument of Asness, Frazzini, and Pedersen (2012), low-risk assets, bonds in particular, offer higher risk-adjusted returns, and an unconstrained investor should lever them. In our data, global bonds have a Sharpe ratio of 0.29, no higher than the 0.31 of stocks (Table I). The gains come instead from managed futures, which have a Sharpe ratio of 0.49 and gain 8.4% on average in the country-years in which stocks lose most. Removing managed futures from either diversified strategy raises its equivalent savings rate, whereas removing bonds lowers it. Gold contributes after 1970, when its price was free.

The choice of retirement outcome also matters in its own right. Under the 4% rule, a strategy with low expected returns can preserve its capital simply because it promises a small income. At 175% exposure, the risk parity strategy has a ruin probability of only 3.8%, but it fails to fund the path-matched income in 14.2% of lifetimes, twice the rate of the all-equity strategy. Levered to equity volatility, at 265% exposure, risk parity does better than the proportional strategy, but this advantage is fragile.

Our conclusions rest on two conditions: cheap financing and the returns of our managed-futures series, which we build from public futures data back to 1927. Both are within reach of individual investors today. At equal volatility, the proportional strategy keeps its saving advantage up to a borrowing spread of about 2.75% and its income advantage up to about 1.3%. Individual investors do not need a margin loan to borrow at such rates. Box spreads on S&P 500 index options, which trade in ordinary brokerage accounts, have been priced at 0.2% to 0.5% above comparable SOFR or Treasury rates (CME Group, 2024; Wang,

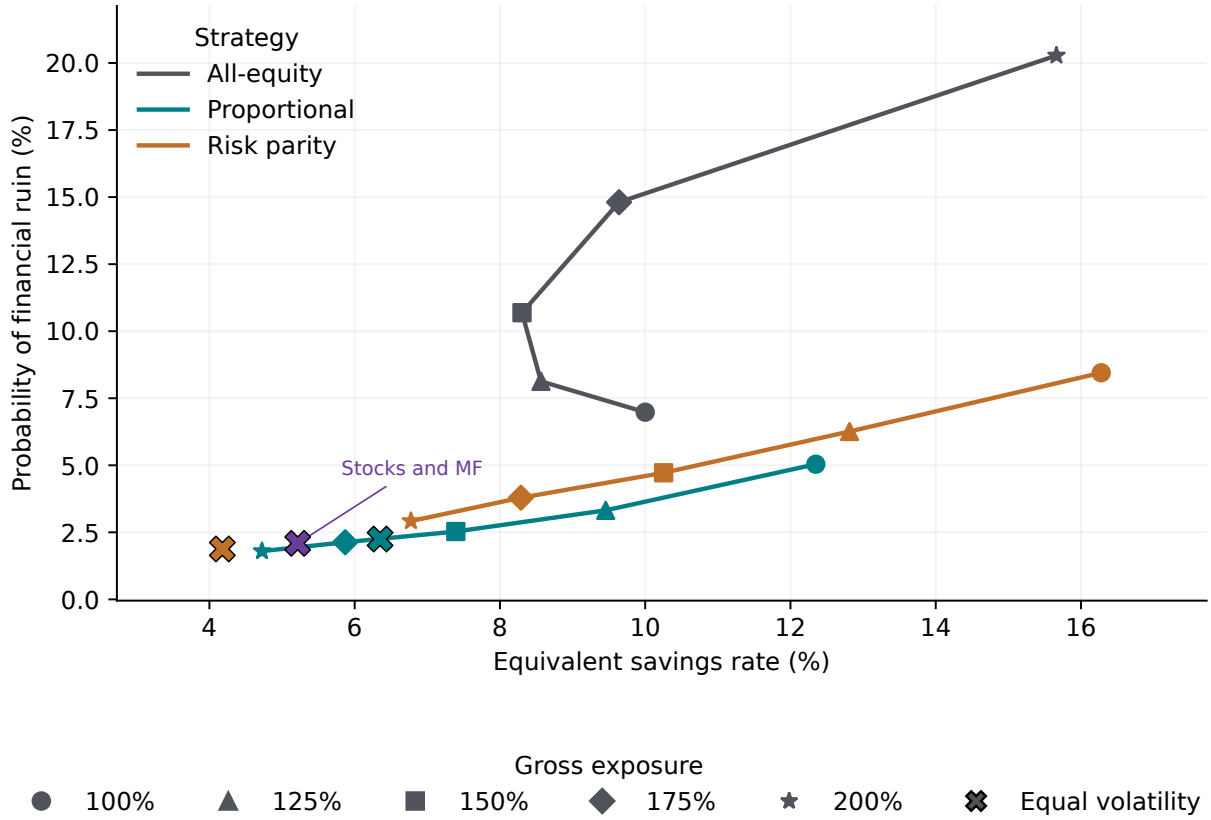


Figure 1. Equivalent savings rates and ruin probabilities. The figure plots the probability of financial ruin against the equivalent savings rate for the all-equity, proportional, and risk parity strategies at gross exposures of 100%, 125%, 150%, 175%, and 200%. Crosses mark the proportional, risk parity, and stocks-and-managed-futures strategies at the volatility of the unlevered all-equity strategy (166%, 265%, and 132% exposure). The equivalent savings rate equates expected utility with that of the all-equity strategy at 100% exposure and a 10% savings rate. Results are based on 10,000 bootstrap simulations from the 1927 to 2025 sample.

2025), and the financing embedded in S&P 500 futures, which return-stacked funds use, cost 0.3% on average over 2021 to 2023 (D. E. Shaw & Co., 2025). Our series earned 0.7 to 2.0 percentage points per year more than commercial trend-following indexes after 2000, yet even with a penalty of 2% per year on its returns, the proportional and stocks-and-managed-futures strategies still improve on the all-equity strategy in utility, ruin, and income funding. Funds that replicate these indexes are available to individual investors, including as UCITS exchange-traded funds (iM Global Partner, 2026). The results also hold across resampled calendar histories, managed-futures signals, samples, preferences, and savings policies.

We contribute to the normative literature on lifecycle investment strategies [e.g., Cocco, Gomes, and Maenhout (2005); Gomes (2020); and Anarkulova, Cederburg, and O’Doherty (2025)]. Our primary contribution is to separate the value of diversification from the value of leverage in a lifecycle model with labor income, Social Security, and longevity risk. Most lifecycle studies restrict households to stocks, bonds, and bills and rule out borrowing.³ We show that the gains from leverage depend on the assets being levered, and that the gains from diversification depend on leverage. Our second contribution is to distinguish capital preservation from the ability to fund a given retirement income. Finally, we use a panel built entirely from public data, and we release the data and code.

2 Leverage and diversification in lifecycle investing

In the models of Merton (1969) and Samuelson (1969), a sufficiently risk-tolerant investor borrows to hold more of the risky asset. Later lifecycle models typically rule out borrowing or make it costly. Cocco, Gomes, and Maenhout (2005) impose borrowing constraints, and Davis, Kubler, and Willen (2006) show that the spread between borrowing and lending rates lowers optimal equity holdings.

A separate literature argues that borrowing constraints distort the choice of risky assets, not only the level of risk. Asness (1996) shows that a levered stock-bond portfolio historically delivered equity-like returns with lower volatility than an all-equity portfolio. Frazzini and Pedersen (2014) and Asness, Frazzini, and Pedersen (2012) provide a theory and evidence: investors who cannot or will not borrow overweight risky assets, which lowers the risk-adjusted returns of those assets. Risk parity portfolios, which equalize risk contributions across asset classes (Maillard, Roncalli, and Teiletche, 2010), are the standard institutional application of this logic. Their historical performance depends on the financing cost and

³Davis, Kubler, and Willen (2006) show that realistic borrowing costs reduce the demand for equity over the life cycle. Ayres and Nalebuff (2010) and Ayres and Nalebuff (2013) propose leveraged equity positions for young investors to diversify across time. At the lowest spread of Anarkulova, Cederburg, and O’Doherty (2025), their couple borrows 100% of wealth and holds 15% in bonds.

the sample period (Anderson, Bianchi, and Goldberg, 2012). Bhansali et al. (forthcoming) show that treating trend following as a separate asset class within risk parity improves its performance, notably in 2022, when stocks and bonds fell together. Return-stacking funds bring levered diversification to retail investors by combining a full exposure to stocks or bonds with futures-based diversifiers (Gordillo and Hoffstein, 2021).

Our diversifying asset classes differ in their economic sources of return. Trend-following strategies earn positive average returns across asset classes and centuries and have historically performed well in extended equity declines (Moskowitz, Ooi, and Pedersen, 2012; Hurst, Ooi, and Pedersen, 2017); their returns are commonly attributed to initial underreaction and delayed overreaction to news, not to leverage aversion. Gold is a store of value whose real price depends on the monetary regime (Erb and Harvey, 2013). Studies of retirement withdrawals find that trend following can reduce sequence-of-returns risk (Clare, Seaton, Smith, and Thomas, 2017).

3 Data

Following Anarkulova, Cederburg, and O’Doherty (2022) and Anarkulova, Cederburg, and O’Doherty (2025), we model forward-looking returns with the historical returns of a broad cross section of developed countries. All inputs to the simulations come from public sources.⁴ The starting point is release 6 of the Jordà-Schularick-Taylor Macrohistory Database (Jordà et al., 2019), which contains annual nominal total returns on stocks and ten-year government bonds, short-term bill rates, consumer price indexes, and exchange rates for 16 developed countries through 2020. We extend each series through 2025 using dividend-adjusted local indexes, locally listed exchange-traded funds, and the Global Macro Database (Müller, Xu, Lehib, and Chen, 2025). The dataset is an unbalanced panel of 1,561 country-years from 1927 to 2025. Section A of the internet appendix provides details on the data sources and return calculations.

All returns for a given country-year are real returns for local investors. The international stock return for a country is the market-capitalization-weighted return on all other markets, converted into local currency, with weights from Kuvshinov and Zimmermann (2022). As in Anarkulova, Cederburg, and O’Doherty (2025), the stock allocation is 33% domestic stocks and 67% international stocks, and we refer to this mix as stocks.

Global bonds are an equally weighted portfolio of the ten-year government bonds of all countries in the panel in a given year, including the investor’s own country. Foreign bond

⁴Proprietary indexes of commercial trend-following funds are used only to validate the managed-futures series.

positions are hedged into local currency with one-period forwards on the initial principal, priced at covered interest parity. We deduct a fund fee and a hedging cost of 0.10% per year each.⁵

Gold returns are based on the annual US dollar price of gold from MeasuringWorth (Officer and Williamson, 2024) through 1967 and the London Bullion Market Association afternoon fixing (London Bullion Market Association, 2025) from 1968, when the private London market was separated from the official price, less a custody cost of 0.40% per year. Before 1968, the dollar price of gold was administered, so early gold returns reflect a different monetary regime (Erb and Harvey, 2013), and we examine the post-1970 period separately.

No investable managed-futures index spans our sample period, so we construct a time-series momentum strategy from monthly futures-equivalent returns following Moskowitz, Ooi, and Pedersen (2012) and Hurst, Ooi, and Pedersen (2017). The signal for each market averages the signs of its prior one-, six-, and twelve-month returns, lagged one month. Markets are scaled to 15% volatility and sectors weighted by inverse volatility; the portfolio is scaled to 10% volatility over the preceding 36 months, with leverage capped at three. The universe expands as data become available across equity indexes, government bonds, commodities, and currencies. The strategy earns US dollar collateral income, and we deduct a management fee of 0.85% per year and trading costs of three basis points per unit traded, or about 0.61% per year. The annual series is hedged into local currency in the same manner as global bonds.

The managed-futures series has important limitations. It omits commodity carry and roll returns, uses synthetic bond futures and price-only equity indexes, and trades a narrow set of markets before 1960; currencies enter only in 1971. Its signal follows studies published after much of our sample, so its design is not free of hindsight. For a US investor, the series earns an average real return of 8.05% per year with a standard deviation of 14.18% over 1927 to 2025 (Table A.VI of the internet appendix). The most direct test of the series is its record against commercial funds. Over 2000 to 2025, it earned 6.11% per year in US dollars, compared with 4.20% for the SG CTA index, 5.38% for the SG Trend index, and 4.09% for the BTOP50 index, and its Sharpe ratio was 0.41, compared with 0.27 to 0.30. The gap comes from 2000 to 2009. Over 2010 to 2025, the series earned 2.44% per year with a Sharpe ratio of 0.14, below all three indexes (2.70% to 3.64% and 0.18 to 0.24), consistent with the weaker trend-following returns documented by Babu et al. (2020). The series is also only moderately correlated with the indexes, at 0.55 to 0.59 over 2000 to 2025. Section 5.5

⁵Liquid forward markets did not exist for many panel currencies before the 1970s, and capital controls restricted cross-border positions under the Bretton Woods system. Hedged returns before then are therefore a hypothetical construction. The covered-parity assumption also ignores deviations such as the cross-currency basis documented by Du, Tepper, and Verdelhan (2018).

reports the reduction in managed-futures returns at which our conclusions change.

Table I summarizes the properties of the four asset classes. Over the full sample, stocks earn the highest average real return (8.12% per year), followed by managed futures (7.37%), gold (3.99%), and global bonds (1.94%). In excess of the local bill return, managed futures have a Sharpe ratio of 0.49, compared with 0.31 for stocks, 0.29 for bonds, and 0.12 for gold. The rankings are the same after 1970. In the country-years in the worst decile of stock returns, stocks lose 25.0% on average, bonds lose 2.1%, gold gains 2.6%, and managed futures gain 8.4%. After 1970, gold and managed futures gain 10.5% and 15.4% in these years. Bonds suffer during inflationary periods, with a ten-year correlation with inflation of -0.78 , similar to the -0.78 reported by Anarkulova, Cederburg, and O’Doherty (2025) for 30-year horizons.⁶

4 Methods

We adopt the lifecycle framework of Anarkulova, Cederburg, and O’Doherty (2025) at an annual frequency. Their analysis shows that a fixed-weight strategy of 33% domestic and 67% international stocks achieves virtually the same expected utility as their optimal age-based strategy, so we focus on fixed-weight strategies throughout.

4.1 Lifecycle design and income

Households are composed of a female and a male of equal age. Each member of the household is eligible to work and save from age 25 to age 64, and the couple retires at age 65. In the base case, individuals save $r_c = 10\%$ of their labor income for retirement, and individuals earning less than $Y_{\min} = \$15,000$ (in 2022 USD) in a given year do not contribute. Contributions are made at the beginning of each year. We model labor income using model 6 of Guvenen, Karahan, Ozkan, and Song (2021), as in Anarkulova, Cederburg, and O’Doherty (2025). Log income contains a quadratic age profile, a persistent component with autoregressive coefficient 0.959, transitory shocks, and age- and income-dependent nonemployment. Both members of the couple have the median parameter values. Social Security benefits are computed from each simulated earnings history using the 2022 benefit formulas, including spousal and survivor benefits, and Supplemental Security Income provides a consumption floor in retirement.

At retirement, the couple withdraws $r_w = 4\%$ of its retirement wealth in the first year and

⁶The annual correlation between gold and stocks in the full sample is 0.60. This correlation reflects country-years with large currency depreciations and high inflation, in which both gold and international stocks rise in local currency. At the ten-year horizon, the correlation is -0.35 .

Table I: Empirical properties of real returns on the four asset classes

The table summarizes the empirical properties of annual real returns for local investors on the four asset classes available to the household. The underlying data are an annual panel of 16 developed countries covering 1927 to 2025, described in Section 3. Returns are net of the managed-futures fee and turnover cost, gold custody, and the currency-hedging cost. Stocks are 33% domestic and 67% international. The Sharpe ratio is the mean annual return in excess of the local real bill return divided by its standard deviation. The ten-year statistics use overlapping ten-year log returns within each country. The last row reports the average return of each asset class in country-years in the lowest decile of stock returns. Panel B restricts the sample to 1970 to 2025.

	Stocks (33/67)	Global bonds	Gold	Managed futures
Panel A: 1927–2025 (1,561 country-years)				
Mean (%)	8.12	1.94	3.99	7.37
Standard deviation (%)	25.28	8.91	31.62	15.77
Sharpe ratio (excess of bills)	0.31	0.29	0.12	0.49
Correlation with stocks (annual)	1.00	0.28	0.60	0.21
Correlation with stocks (10-year)	1.00	0.40	−0.35	0.15
Correlation with inflation (annual)	−0.12	−0.47	−0.08	−0.24
Correlation with inflation (10-year)	−0.23	−0.78	−0.08	−0.29
Mean in worst stock decile (%)	−24.95	−2.12	2.63	8.44
Panel B: 1970–2025 (894 country-years)				
Mean (%)	7.79	2.75	6.07	8.01
Standard deviation (%)	17.24	7.97	22.48	12.25
Sharpe ratio (excess of bills)	0.42	0.30	0.23	0.61
Correlation with stocks (annual)	1.00	0.28	−0.04	−0.06
Correlation with stocks (10-year)	1.00	0.53	−0.64	−0.03
Correlation with inflation (annual)	−0.22	−0.38	0.03	0.11
Correlation with inflation (10-year)	−0.15	−0.52	0.07	0.39
Mean in worst stock decile (%)	−26.81	−0.08	10.47	15.36

the same real amount in each subsequent year (Bengen, 1994). The fixed real withdrawal makes retirement outcomes sensitive to the order of returns (Estrada, 2018). Mortality follows sex-specific Gompertz-Makeham laws calibrated to the longevity moments reported by Anarkulova, Cederburg, and O’Doherty (2025) from Social Security Administration tables (Social Security Administration, 2020), and the lifespan of each individual is random.⁷

4.2 Household utility and allocation strategies

Household utility is determined by annual retirement consumption and a bequest:

$$U(C, B) = \sum_{t=65}^{T_{\max}} \frac{(C_t/\sqrt{H_t})^{1-\gamma}}{1-\gamma} + \theta \frac{(B+k)^{1-\gamma}}{1-\gamma}, \quad (1)$$

where C_t is real consumption at age t , H_t is the household’s size, B is the real bequest, and $T_{\max}$ is the age at death of the last survivor. Following Anarkulova, Cederburg, and O’Doherty (2025), we use the estimates of De Nardi, French, and Jones (2010): a coefficient of relative risk aversion of $\gamma = 3.84$, a bequest intensity of $\theta = 2,360$, and a bequest curvature parameter of $k = \$490,000$. Anarkulova, Cederburg, and O’Doherty (2025) evaluate monthly consumption and multiply the annual bequest intensity by 12^γ . With uniform consumption within the year, our annual specification preserves the relative weight of consumption and bequest in their utility function.

The household invests in stocks, global bonds, gold, and managed futures and can borrow at the local bill rate plus a spread. Let w be the vector of asset class weights as fractions of wealth, and let $G = \mathbf{1}'w$ be gross exposure. The real portfolio return in year t is

$$R_t^p = w'R_t + (1-G)R_t^f - \max(G-1, 0)\phi - h\kappa, \quad (2)$$

where R_t is the vector of real asset class returns, R_t^f is the local real bill return, ϕ is the borrowing spread, h is the weight of bonds and managed futures, and κ is the hedging cost. In the base case, $\phi = 0.30\%$, which is close to the lowest spread of 0.37% considered by Anarkulova, Cederburg, and O’Doherty (2025) and estimated by Binsbergen, Diamond, and Grotteria (2022) from derivative prices. This is the cost of financing through box spreads on index options (Wang, 2025) or through the futures held by return-stacked funds, both open to individual investors (Section 6). It is not the rate of a retail margin loan, which we cover

⁷As in Anarkulova, Cederburg, and O’Doherty (2025), the model combines US earnings, public pensions, and mortality with returns pooled across developed countries. The simulations therefore describe a US household exposed to the return histories of other countries. They do not model the pension systems of those countries.

with higher spreads.

We consider three families of strategies, each evaluated at gross exposures of $G = 100\%$, 125% , 150% , 175% , and 200% :

- the *all-equity strategy*, which invests G in stocks;
- the *proportional strategy*, which invests $0.40G$ in stocks, $(4/15)G$ in global bonds, $G/6$ in gold, and $G/6$ in managed futures; and
- the *risk parity strategy*, which invests $0.1453G$ in stocks, $0.4747G$ in global bonds, $0.1205G$ in gold, and $0.2595G$ in managed futures.

The proportional weights start from a portfolio of 60% stocks and 40% bonds and borrow 50% of wealth to add 25% gold and 25% managed futures; the resulting proportions are then held fixed as gross exposure varies. This practitioner rule is similar to the return-stacking portfolios offered to retail investors, and its weights were fixed before we examined household outcomes. We keep it as our main diversified strategy even though bonds turn out to lower utility, because dropping an asset class after observing its contribution would select on outcomes. We also report a strategy that holds only stocks and managed futures in the same relative proportions, and the weights that maximize expected utility in our sample. The risk parity weights equalize the contributions of the four asset classes to the variance of an unlevered portfolio (Maillard, Roncalli, and Teiletche, 2010), using the sample covariance matrix of annual real returns over all 1,561 country-years. This estimation uses the full sample and therefore gives the risk parity strategy an informational advantage.

Gross exposure is not a measure of risk: at 175%, the three families have annual volatilities of 44.0%, 26.6%, and 17.1%. Our main comparison therefore scales each diversified strategy to the 25.28% annual volatility of the unlevered all-equity strategy in the pooled panel, the comparison implied by the leverage-aversion argument (Asness, 2024). This gives exposures of 166% for the proportional strategy, 265% for risk parity, and 132% for stocks and managed futures, which we hold fixed in every sensitivity analysis.

Gross exposure G is the sum of the allocations to the four asset classes as fractions of wealth, and $G - 1$ is borrowing. It is not the notional value of the underlying contracts. The managed-futures series is itself a futures strategy that targets 10% volatility and can hold up to three times its capital in notional positions, and bonds and foreign returns are hedged with forwards. A 28% allocation to managed futures counts as 28% of wealth whatever its internal notional. The borrowed part of the portfolio requires collateral and can be liquidated after losses; Section 5.5 reports margin tests and Section 6 discusses financing.

4.3 Simulation procedure

We simulate 10,000 lifecycle outcomes. Each draw combines random lifespans for the two members of the couple, simulated labor incomes, and a time series of annual real asset class returns from a stationary block bootstrap in the spirit of Politis and Romano (1994). We select a random starting country-year and draw a block of consecutive years from the same country, with a geometric block length of ten years on average; each country-year supplies all asset class returns jointly. If a country’s data end before the block is complete, the block continues at the start of a randomly drawn country’s sample, without a currency conversion. Wealth evolves as $W_{t+1} = \max\{0, (W_t + S_t)(1 + R_{t+1}^p)\}$ from $W_{25} = 0$, where S_t is the contribution and R^p is given by equation (2). In retirement, the couple withdraws $D_t = \min\{0.04 W_{65}, W_t\}$ at the beginning of each year and consumes $C_t = \max\{D_t + SS_t, SSI_t\}$, where SS_t is the Social Security benefit and SSI_t is Supplemental Security Income. The bequest is the wealth left at the death of the last survivor. Expected utility is the average of equation (1) across draws. All strategies share the same return, income, and mortality draws, and we find equivalent savings rates by bisection. Portfolios are rebalanced to their target weights once a year.

4.4 Three measures of retirement outcomes

We compare strategies with three measures, summarized in Table II. The first is the equivalent savings rate of Anarkulova, Cederburg, and O’Doherty (2025), the savings rate r_c^* that gives a strategy the same expected utility as the all-equity strategy with 100% exposure and a 10% savings rate:

$$E[U_j \mid r_c^*] = E[U_{\text{All-equity}} \mid r_c = 10\%]. \quad (3)$$

A rate below 10% means that the strategy provides higher expected utility. It measures utility, not income: a household saving r_c^* has the same expected utility as the all-equity household, not the same retirement income on each path.

The second is the probability of financial ruin, defined as exhausting retirement wealth under the 4% rule before the death of the last survivor. Contributions, retirement wealth, and the withdrawal all scale with the savings rate, so this probability does not depend on the savings rate (Section B.4 of the internet appendix). A strategy with a low expected return can therefore have a low ruin probability because it withdraws a small income.

The third measures the ability to fund a given income. Each household withdraws 4% of the retirement wealth that the all-equity strategy with 100% exposure and a 10% savings rate accumulates on the same simulated path, whatever strategy it follows. We call this withdrawal the path-matched income: it differs across paths and is high when equities have

done well. The path-matched income failure is the probability that the strategy’s wealth cannot pay it before the death of the last survivor. We compute it with a 10% savings rate and, in Table II, also at the strategy’s own equivalent savings rate.

Table II: Three measures of retirement outcomes

The table reports three measures for the strategies at the volatility of the unlevered all-equity strategy. (1) The equivalent savings rate gives the strategy the expected utility of the all-equity strategy saving 10%. (2) The ruin probability applies the 4% rule to the strategy’s own retirement wealth; it does not depend on the savings rate. (3) The path-matched income failure is the probability of not paying 4% of the retirement wealth that the all-equity strategy saving 10% accumulates on the same path, when the strategy saves 10% or its own equivalent savings rate. Each measure is based on the same 10,000 simulations.

Strategy	(1)	(2)	(3) Path-matched income failure	
	Equivalent savings rate	Ruin probability	Saving 10%	Saving rate (1)
All-equity 100%	10.00%	6.98%	6.98%	6.98%
Proportional 166%	6.35%	2.25%	3.18%	13.10%
Risk parity 265%	4.18%	1.88%	2.17%	16.86%
Stocks and MF 132%	5.22%	2.09%	0.82%	5.87%

5 Results

5.1 Levered diversification

Table III reports the asset class weights, return volatility, ruin probability, and equivalent savings rate for each strategy. Our annual implementation closely matches Anarkulova, Cederburg, and O’Doherty (2025): the unlevered all-equity strategy has a ruin probability of 7.0%, as in their study, and their ruin probabilities of 17.1% for domestic stocks and 16.9% for a 60/40 strategy compare with 15.5% and 13.7% in our data, although their data cover 39 countries from 1890.

Diversification and leverage improve expected utility only in combination. Panel B shows that the unlevered proportional strategy lowers the ruin probability from 7.0% to 5.0% but raises the equivalent savings rate to 12.35%: it reduces risk but gives up too much expected return. Panel A shows that leverage without diversification provides small utility gains at a large cost in capital preservation. The equivalent savings rate of the all-equity strategy falls to 8.56% at 125% exposure and 8.30% at 150%, but rises to 9.64% at 175% and 15.66% at 200%, while its ruin probability rises at every step, from 7.0% to 20.3%.

Panel E reports the strategies at equal volatility. The proportional strategy at 166%

Table III: Economic value of levered allocation strategies

The table reports the asset class weights, annual return volatility, probability of financial ruin, and equivalent savings rate for the all-equity strategy (Panel A), the proportional strategy (Panel B), and the risk parity strategy (Panel C) at gross exposures from 100% to 200% of wealth. Panel D removes global bonds from the proportional strategy and redistributes their weight in proportion to the remaining weights (stocks, gold, and managed futures) or to stocks and managed futures only, at 175% exposure. Panel E scales each diversified strategy to the 25.28% volatility of the unlevered all-equity strategy. In Panels D and E, the first column names the strategy, and gross exposure is 100% plus borrowing. Stocks are 33% domestic and 67% international. Borrowing is financed at the local bill rate plus 0.30%. Volatility is the standard deviation of annual real portfolio returns in the pooled panel. Each comparison is based on 10,000 bootstrap simulations of household outcomes that share return, earnings, and mortality draws. The household saves 10% of labor income. The probability of financial ruin is the probability of exhausting wealth before the death of the last survivor under the 4% rule. The equivalent savings rate is the savings rate that equates expected utility from retirement consumption and bequest with that of the all-equity strategy at 100% exposure and a 10% savings rate.

Gross exposure	Asset class weights (% of wealth)					Volatility	Ruin probability	Equivalent savings rate
	Stocks	Global bonds	Gold	Managed futures	Borrowing			
Panel A: All-equity strategies								
100%	100%	0%	0%	0%	0%	25.28%	6.98%	10.00%
125%	125%	0%	0%	0%	25%	31.47%	8.13%	8.56%
150%	150%	0%	0%	0%	50%	37.71%	10.69%	8.30%
175%	175%	0%	0%	0%	75%	43.99%	14.81%	9.64%
200%	200%	0%	0%	0%	100%	50.28%	20.28%	15.66%
Panel B: Proportional strategies								
100%	40%	27%	17%	17%	0%	15.66%	5.04%	12.35%
125%	50%	33%	21%	21%	25%	19.23%	3.32%	9.46%
150%	60%	40%	25%	25%	50%	22.88%	2.53%	7.39%
175%	70%	47%	29%	29%	75%	26.58%	2.13%	5.87%
200%	80%	53%	33%	33%	100%	30.31%	1.80%	4.72%
Panel C: Risk parity strategies								
100%	15%	47%	12%	26%	0%	10.82%	8.45%	16.28%
125%	18%	59%	15%	32%	25%	12.83%	6.26%	12.81%
150%	22%	71%	18%	39%	50%	14.95%	4.72%	10.25%
175%	25%	83%	21%	45%	75%	17.13%	3.79%	8.29%
200%	29%	95%	24%	52%	100%	19.36%	2.92%	6.77%
Panel D: Strategies without bonds								
Stocks/gold/MF	95%	0%	40%	40%	75%	34.95%	1.92%	4.15%
Stocks/MF	124%	0%	0%	51%	75%	33.30%	2.05%	3.25%
Panel E: Strategies at the volatility of the unlevered all-equity strategy								
Proportional	67%	44%	28%	28%	66%	25.28%	2.25%	6.35%
Risk parity	38%	126%	32%	69%	165%	25.28%	1.88%	4.18%
Stocks/gold/MF	69%	0%	29%	29%	26%	25.28%	2.43%	6.89%
Stocks/MF	93%	0%	0%	39%	32%	25.28%	2.09%	5.22%

exposure has an equivalent savings rate of 6.35% and a ruin probability of 2.3%; the 95% confidence interval for its difference in ruin probability with the all-equity strategy is -5.2 to -4.3 percentage points. Its returns are as volatile as those of the all-equity strategy but much less concentrated in stock market outcomes. On the exposure grid, the gains increase with leverage up to 200%, where the equivalent savings rate is 4.72% and the ruin probability 1.8%, at a volatility of 30.3%, above that of the all-equity strategy. The stocks-and-managed-futures strategy reaches equity volatility at only 132% exposure, with an equivalent savings rate of 5.22% and a ruin probability of 2.1%.

Risk parity illustrates why the basis of comparison matters. At 175%, where its volatility is only 17.1%, it does worse than the proportional strategy (8.29% equivalent savings rate, 3.8% ruin). At equal volatility, it needs 265% exposure and does better (4.18% and 1.9%). It attains this result by borrowing 165% of wealth and by holding 69% of wealth in managed futures, and Section 5.5 shows that its advantage is fragile.

5.2 Capital preservation and income funding

Table IV and Figure 2 compare the probability of financial ruin with the path-matched income failure on the exposure grid, with a 10% savings rate. The two outcomes diverge sharply for strategies with low expected returns. The unlevered proportional strategy has a lower ruin probability than the all-equity strategy (5.0% versus 7.0%), but it fails to fund the path-matched income in 31.5% of lifetimes. The unlevered risk parity strategy fails in 47.4%. These strategies preserve capital by accumulating less of it. The levered all-equity strategy shows the opposite pattern: at 175% exposure, its ruin probability is 14.8%, but it fails to fund the path-matched income in only 6.3% of lifetimes.

At 175% exposure, the risk parity strategy has a ruin probability of 3.8% but a 14.2% probability of failing to fund the path-matched income. With nearly half of its unlevered weight in bonds, it produces stable but small retirement accounts. At equal volatility, the three diversified strategies fund the path-matched income more reliably than the all-equity strategy when they save 10%, with failure probabilities of 3.2% for the proportional strategy, 2.2% for risk parity, and 0.8% for stocks and managed futures (Table II).

The comparison changes when each strategy saves only its equivalent savings rate. The proportional strategy then fails to fund the path-matched income in 13.1% of lifetimes and risk parity in 16.9%, compared with 7.0% for the all-equity strategy. About 70% of these failures occur in paths where the all-equity strategy accumulates more than its median wealth, so that the income to fund is high: the diversified strategies give up part of the upside of equity booms in exchange for protection in equity crises, which is what the utility

Table IV: Capital preservation and funding the path-matched income

The table compares two retirement outcomes for the all-equity (Panel A), proportional (Panel B), and risk parity (Panel C) strategies. The probability of financial ruin applies the 4% rule to each strategy's own wealth at retirement. The probability of failing to fund the path-matched income instead fixes each household's real withdrawal at 4% of the wealth that the all-equity strategy at 100% exposure accumulates on the same simulated path, so the income to fund differs across paths. Retired households are those that reach retirement with a positive planned withdrawal. Each comparison is based on 10,000 bootstrap simulations of household outcomes that share return, earnings, and mortality draws. The household saves 10% of labor income. The last column reports the 95% confidence interval for the paired difference in failure probability relative to the all-equity strategy at 100% exposure, in percentage points, and reflects Monte Carlo error only.

Gross exposure	Probability of financial ruin	Failure to fund path-matched income		95% interval for difference (pp)
		All households	Retired households	
Panel A: All-equity strategies				
100%	6.98%	6.98%	7.10%	—
125%	8.13%	3.83%	3.90%	[−3.52, −2.78]
150%	10.69%	4.18%	4.25%	[−3.24, −2.36]
175%	14.81%	6.25%	6.36%	[−1.25, −0.21]
200%	20.28%	11.46%	11.66%	[3.83, 5.13]
Panel B: Proportional strategies				
100%	5.04%	31.49%	32.03%	[23.60, 25.42]
125%	3.32%	15.04%	15.30%	[7.34, 8.78]
150%	2.53%	6.00%	6.10%	[−1.54, −0.42]
175%	2.13%	2.17%	2.21%	[−5.29, −4.33]
200%	1.80%	1.03%	1.05%	[−6.43, −5.47]
Panel C: Risk parity strategies				
100%	8.45%	47.39%	48.20%	[39.40, 41.42]
125%	6.26%	33.93%	34.51%	[25.99, 27.91]
150%	4.72%	22.67%	23.06%	[14.83, 16.55]
175%	3.79%	14.25%	14.49%	[6.52, 8.02]
200%	2.92%	8.67%	8.82%	[1.03, 2.35]

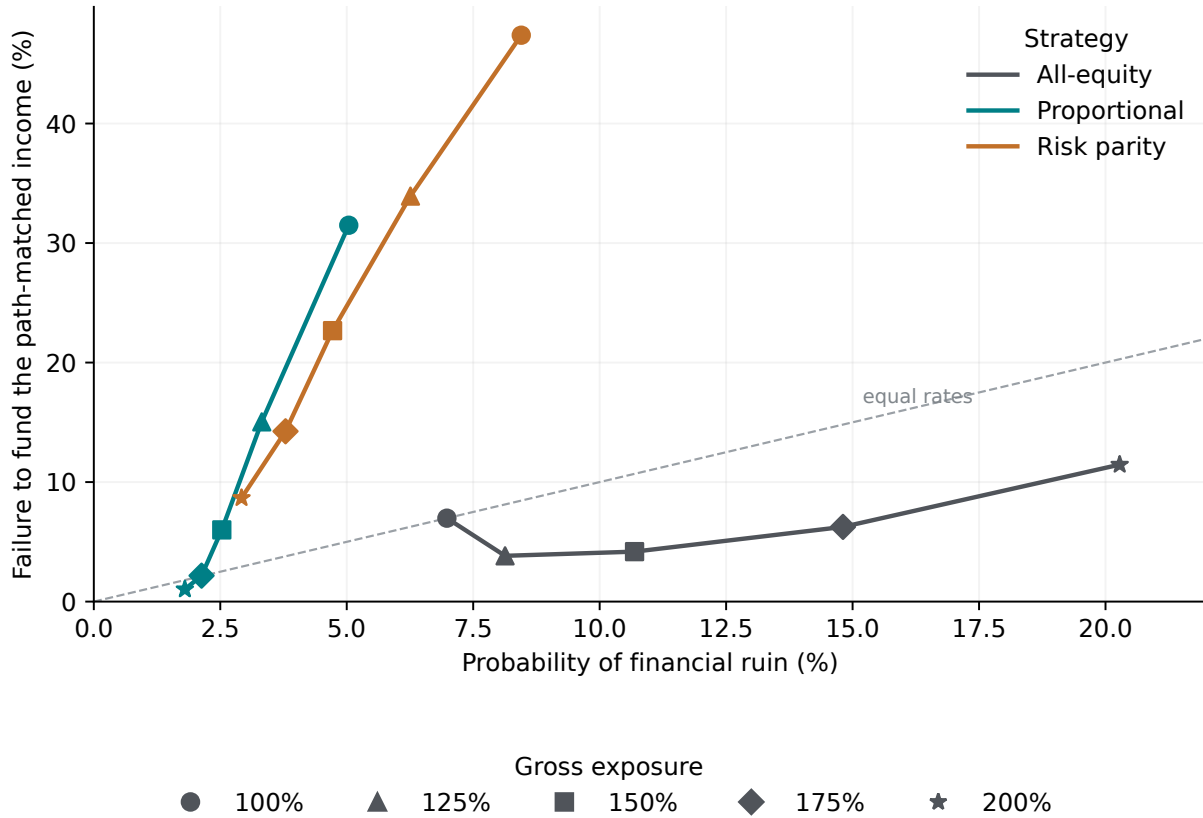


Figure 2. Capital preservation and income funding. The figure plots the probability of failing to fund the path-matched income against the probability of financial ruin for the all-equity, proportional, and risk parity strategies at gross exposures from 100% to 200%. The probability of financial ruin applies the 4% rule to each strategy's own retirement wealth. The path-matched income is 4% of the retirement wealth of the all-equity strategy at 100% exposure on the same simulated path. Strategies above the dashed line fund the path-matched income less reliably than they preserve their own capital. Results are based on 10,000 bootstrap simulations from the 1927 to 2025 sample with a 10% savings rate.

function rewards. Only the stocks-and-managed-futures strategy funds the path-matched income more reliably than the all-equity strategy at its own equivalent savings rate (5.9% failure). Capital preservation, utility, and income funding are therefore distinct objectives, and a low ruin probability can reflect a low planned income rather than a safe portfolio.

5.3 Sources of the gains

The gains come mainly from managed futures, and bonds lower utility when they replace stocks. Table V reports changes in the equivalent savings rate at 175% exposure, the grid point closest to equal volatility for the proportional strategy, in the full sample and after 1970. Panel A adds each diversifying asset class to the levered all-equity strategy at its proportional-strategy weight, financed by reducing stocks. Panels B and C remove each asset class from the diversified strategies and reallocate its weight to stocks. Table C.I of the internet appendix reports ruin and path-matched income outcomes in the full sample.

Managed futures are indispensable. Adding them to the levered all-equity strategy lowers the equivalent savings rate by 5.4 percentage points in the full sample and 8.3 points after 1970. Removing them raises the equivalent savings rate of both diversified strategies in both samples, and, with their weight spread pro rata (Table C.I), ruin from 2.1% to 3.7% (proportional) and 3.8% to 8.1% (risk parity).

In our panel and at a given exposure, global bonds lower utility. Removing them lowers the equivalent savings rate of the proportional strategy by 1.7 and 1.0 points in the two samples and, spread pro rata, its path-matched income failure from 2.2% to 0.9%. Panel D of Table III reports two bond-free versions of the proportional strategy at 175%: with stocks, gold, and managed futures, the equivalent savings rate is 4.15%, and with stocks and managed futures only, 3.25%. These results are consistent with the long-horizon properties of bonds documented by Anarkulova, Cederburg, and O’Doherty (2025). They concern hedged ten-year government bonds in a panel that includes the inflations of the 1940s and 1970s, and bonds can still add value indirectly, by lowering volatility and so permitting more leverage on the other asset classes, as in the risk parity strategy at equal volatility.

Gold’s contribution depends on the monetary regime. In the full sample, which includes four decades of administered gold prices, removing gold from the proportional strategy lowers its equivalent savings rate by 0.8 points. After 1970, adding gold to the levered all-equity strategy lowers the equivalent savings rate by 7.3 points, nearly as much as managed futures, and removing it from the proportional strategy raises the rate by 0.4 points. Once managed futures are present, gold adds little, as the two asset classes pay off in the same episodes.

Panel D of Table V reports the weights that maximize expected utility at 175% exposure,

Table V: Sources of the gains from levered diversification

The table reports the change in the equivalent savings rate, in percentage points, from changing the asset class weights of a strategy at 175% gross exposure. A negative change means that the household needs to save less to attain the expected utility of the all-equity strategy at 100% exposure and a 10% savings rate. Panel A adds global bonds, gold, managed futures, or all three to the levered all-equity strategy at their proportional-strategy weights, financed by reducing stocks. Panels B and C remove one asset class from the proportional or risk parity strategy and reallocate its weight to stocks. Reallocation is the weight moved, as a percentage of wealth. Panel D reports the weights that maximize expected utility at 175% exposure in each sample, measured relative to the all-equity strategy at 175%; the last two rows reoptimize after subtracting 3% and 6% per year from managed-futures returns. Each comparison is based on 10,000 bootstrap simulations. The last column draws returns from 1970 to 2025, after the 1968 separation of the free London gold market from the official price. In the full sample, the optimal weights are 83% stocks, 92% managed futures; 113% stocks, 5% gold, 56% managed futures with a 3% haircut; 127% stocks, 2% global bonds, 35% gold, 11% managed futures with a 6% haircut. In the post-1970 sample, they are 72% stocks, 18% gold, 85% managed futures; 96% stocks, 30% gold, 49% managed futures with a 3% haircut; 110% stocks, 4% global bonds, 43% gold, 17% managed futures with a 6% haircut.

Change in weights	Reallocation	Full sample (1927–2025)	Post-1970 sample (1970–2025)
Panel A: Add one asset class to the all-equity strategy at 175%			
+ Global bonds	46.7	−2.1	−4.7
+ Gold	29.2	−3.3	−7.3
+ Managed futures	29.2	−5.4	−8.3
+ All three asset classes	105.0	−3.8	−8.3
Panel B: Remove one asset class from the proportional strategy at 175%			
− Global bonds	46.7	−1.7	−1.0
− Gold	29.2	−0.8	+0.4
− Managed futures	29.2	+1.2	+1.1
− Bonds and gold	75.8	−1.6	+0.0
− All three asset classes	105.0	+3.8	+8.3
Panel C: Remove one asset class from the risk parity strategy at 175%			
− Global bonds	83.1	−4.7	−3.0
− Gold	21.1	−1.4	−0.1
− Managed futures	45.4	+0.9	+1.1
− Bonds and gold	104.2	−4.9	−2.6
− All three asset classes	149.6	+1.4	+6.8
Panel D: Utility-maximizing weights at 175% exposure			
Optimal mix	92.2	−6.9	−10.2
Optimal mix, MF −300bp	61.7	−4.6	−8.6
Optimal mix, MF −600bp	47.6	−3.5	−8.0

an in-sample upper bound on the gains. In the full sample, the optimal strategy lowers the equivalent savings rate by 6.9 points relative to the levered all-equity strategy, compared with 3.8 points for the proportional strategy. The optimal strategy allocates 92% of wealth to managed futures and none to bonds or gold. When managed-futures returns are reduced by 3% or 6% per year, the optimal weights shift toward stocks and gold, and the gain shrinks to 4.6 and 3.5 points. The proportional rule captures about half of the in-sample gain, and removing bonds or gold would improve it, which suggests that it was not tuned to our sample.⁸

5.4 Historical uncertainty and sample periods

The 1,561 country-years of the panel are not 1,561 independent observations. Gold and managed futures are global series, common to all countries in a year up to currency conversion, and stock and bond returns are correlated across countries. For these asset classes, the independent information is closer to the 99 calendar years of the sample. The confidence intervals of Section 5.1 measure only Monte Carlo error given the panel. To measure uncertainty about the history itself, we resample blocks of complete calendar-year cross-sections, with average lengths of 5, 10, or 20 years, and repeat the simulation on each of 100 resulting histories. Table VI reports the results at equal volatility. The proportional strategy has a lower ruin probability than the all-equity strategy in 97% to 100% of the histories, a lower equivalent savings rate in 96% to 99%, and a lower path-matched income failure in 78% to 83%. The stocks-and-managed-futures strategy does better on all three measures in 99% to 100% of the histories.

Panel B of Table VII restricts the sample to 1970 to 2025. The ruin probabilities of the diversified strategies fall to 0.5% or less, compared with 6.7% for the all-equity strategy. These results require caution. The sample begins just before 1973 and 1974, the strongest joint episode for gold and trend following in our data (Table A.VI), and excludes the wartime inflations and currency collapses of the 1930s and 1940s. We therefore interpret the post-1970 results as evidence for the period of free gold markets and broader managed-futures coverage, not as an out-of-sample test.

Panel B also excludes Italy in 1942, when the lira moves from 19.7 to 225.5 per US dollar in the Macrohistory Database, giving an Italian investor a real return of about 1,000% on world stocks, by far the largest observation in the panel. The results are essentially unchanged. Restricting the proportional strategy to asset classes that were investable at each date, with

⁸Section C.2 of the internet appendix decomposes these changes into a mean component and a covariance component with the marginal utility of return. For small changes in weights, the covariance component accounts for most of the value of adding each asset class to the levered all-equity strategy.

Table VI: Retirement outcomes across resampled calendar histories

The table compares each strategy, at the volatility of the unlevered all-equity strategy, with the all-equity strategy at 100% exposure across 100 alternative histories of the panel. Each history resamples blocks of complete calendar-year cross-sections of the 16-country panel with a stationary block bootstrap of the indicated average length and supports 1,000 simulations of household outcomes. The table reports the median and, in brackets, the 5th and 95th percentiles across histories of the difference in ruin probability, the equivalent savings rate, and the difference in path-matched income failure with a 10% savings rate. The last column reports the percentage of histories in which the strategy does better than the all-equity strategy on each of the three measures.

Block (years)	Ruin difference (pp)	Equivalent savings rate (%)	Path-matched income difference (pp)	Histories better (%)
Panel A: Proportional at 166% exposure				
5	−5.40 [−17.52, −0.40]	6.19 [3.77, 9.88]	−5.05 [−19.06, 4.92]	97 / 96 / 78
10	−5.05 [−12.80, −1.18]	6.19 [4.28, 9.27]	−4.75 [−14.33, 2.93]	99 / 97 / 83
20	−4.90 [−10.82, −1.29]	6.43 [4.38, 9.28]	−3.40 [−10.51, 3.44]	100 / 99 / 83
Panel B: Risk parity at 265% exposure				
5	−5.95 [−18.68, −0.40]	4.05 [1.79, 9.26]	−5.65 [−19.93, 7.95]	96 / 96 / 80
10	−5.50 [−14.10, −0.80]	4.04 [2.03, 7.87]	−5.25 [−15.81, 5.70]	99 / 99 / 85
20	−5.15 [−11.66, −1.59]	4.33 [2.29, 7.64]	−3.85 [−11.51, 4.15]	100 / 100 / 85
Panel C: Stocks and MF at 132% exposure				
5	−5.40 [−15.12, −0.80]	5.38 [3.73, 6.98]	−7.60 [−18.93, −1.60]	99 / 100 / 100
10	−4.90 [−11.44, −1.90]	5.45 [3.85, 6.64]	−7.05 [−15.50, −2.90]	100 / 100 / 100
20	−5.15 [−10.40, −1.70]	5.35 [4.07, 6.70]	−6.45 [−12.07, −2.50]	100 / 100 / 100

no gold before 1975 and managed-futures sectors added as their futures markets opened, gives a ruin probability of 2.3% and an equivalent savings rate of 5.77% at 175% exposure (Section B.7 of the internet appendix).

5.5 Borrowing costs, managed-futures returns, and margin requirements

Table VII reports the three measures at equal volatility under alternative assumptions.

5.5.1 Borrowing costs

Panel C raises the borrowing spread from 0.30% to 1.00%, 2.00%, and 3.00%. The equivalent savings rate of the proportional strategy rises to 7.21%, 8.66%, and 10.45%. By linear interpolation, it loses its saving advantage over the unlevered all-equity strategy at a spread of about 2.75%, and its advantage in funding the path-matched income at about 1.3%. At the medium spread of 1.40% of Anarkulova, Cederburg, and O’Doherty (2025), which equals the lowest broker margin rate in April 2024, its interpolated equivalent savings rate is 7.79%. Households need not pay margin rates: box spreads on S&P 500 options have traded at a median of 0.20% to 0.31% above Term SOFR (CME Group, 2024), and the S&P 500 financing spread embedded in futures averaged 0.3% over 2021 to 2023, although it peaked above 1.4% in December 2024 (D. E. Shaw & Co., 2025). Risk parity, which borrows most, loses its saving advantage at a spread of about 2.1%. The stocks-and-managed-futures strategy, which borrows only 32% of wealth, keeps all three advantages at a 3.00% spread, with an equivalent savings rate of 6.62% and a path-matched income failure of 1.7%.

5.5.2 Managed-futures returns

Panel D subtracts an additional 2%, 3%, or 6% per year from managed-futures returns, beyond fees and trading costs. With a 2% reduction, the proportional strategy has an equivalent savings rate of 7.39%, a ruin probability of 2.8%, and a path-matched income failure of 6.0%, all better than the all-equity strategy. It loses its saving advantage at a reduction of about 5.9% per year and its advantage in funding the path-matched income at about 2.5%. The stocks-and-managed-futures strategy keeps its saving advantage up to about 5.9% and its income advantage up to about 4.7%. Risk parity is the most sensitive, losing its income advantage at about 1.5%.

Panel E rebuilds the managed-futures series with other signals: a 1/3/12-month blend, the 12-month signal alone, the 1-month signal alone, and the baseline blend with a fee of 1.70% instead of 0.85%. The alternative series have annual correlations of 0.76 to 0.96 with

Table VII: Retirement outcomes at equal volatility under alternative assumptions

The table reports retirement outcomes, in percent, for the proportional, risk parity, and stocks-and-managed-futures strategies at the gross exposures that give them the volatility of the unlevered all-equity strategy in the base case (166%, 265%, and 132%), held fixed across specifications. For each strategy, the table reports the equivalent savings rate (Equiv.), the probability of financial ruin (Ruin), and the probability of failing to fund the path-matched income, 4% of the retirement wealth of the all-equity strategy at 100% exposure on the same path, with a 10% savings rate (Match). The second column reports the ruin probability of the all-equity strategy, the utility reference in each specification. Panel B draws returns from 1970 to 2025 or excludes Italy in 1942 from the resident panel and the international baskets. Panel C raises the borrowing spread above the local bill rate from 0.30%. Panel D subtracts an additional annual return from managed futures. Panel E rebuilds the managed-futures series with other signals or a doubled fee. Each specification is based on 10,000 bootstrap simulations.

Description	All-equity	Proportional			Risk parity			Stocks and MF		
		Equiv.	Ruin	Match	Equiv.	Ruin	Match	Equiv.	Ruin	Match
Panel A: Base case										
Base case	6.98	6.35	2.25	3.18	4.18	1.88	2.17	5.22	2.09	0.82
Panel B: Alternative samples										
Post-1970 sample	6.67	4.56	0.04	0.28	2.57	0.00	0.09	4.73	0.50	0.06
Excluding Italy 1942	7.19	6.28	2.14	3.05	4.02	1.73	2.08	5.23	2.40	0.76
Panel C: Borrowing spread										
Spread of 1.00%	6.98	7.21	2.62	5.51	5.75	3.06	6.44	5.55	2.31	0.99
Spread of 2.00%	6.98	8.66	3.56	10.87	9.28	6.65	18.40	6.06	2.64	1.23
Spread of 3.00%	6.98	10.45	5.10	18.98	15.76	12.78	37.48	6.62	3.06	1.69
Panel D: Managed-futures return haircut										
Haircut of 2.00%	6.98	7.39	2.75	6.04	6.12	3.29	7.67	6.45	2.94	1.53
Haircut of 3.00%	6.98	7.98	3.16	8.04	7.45	4.58	11.97	7.19	3.47	2.40
Haircut of 6.00%	6.98	10.08	4.80	17.17	14.10	11.20	33.22	10.08	6.19	10.67
Panel E: Managed-futures construction										
1/3/12-month signal	6.98	6.29	2.16	2.72	4.05	1.71	1.36	5.17	2.08	0.81
12-month signal	6.98	6.47	2.43	3.84	4.45	2.46	3.81	5.36	2.44	0.97
1-month signal	6.98	7.16	2.32	5.14	5.67	2.27	5.61	6.17	2.55	1.29
Fee of 1.70%	6.98	6.80	2.39	4.15	4.95	2.31	3.99	5.75	2.40	1.08

the baseline. The proportional strategy’s equivalent savings rate ranges from 6.29% to 7.16%, its ruin probability from 2.2% to 2.4%, and its path-matched income failure from 2.7% to 5.1%, all better than the all-equity strategy. The results do not depend on the signal.

5.5.3 Margin requirements and other design choices

Our model is annual, and it does not simulate margin calls within the year. In the simulations, the proportional strategy at 200% exposure never breaches a 25% maintenance requirement at year-end, whereas the all-equity strategy breaches it in 2.1% of account-years at 200% (Table C.IX). In monthly US data from 1970 to 2025, the proportional strategy never breaches a 25% or 40% requirement at month-end at any exposure up to 200%, while the all-equity strategy breaches the 40% requirement in 13 years at 200% (Table C.X). We have not tested the risk parity strategy at 265% exposure. The internet appendix also shows that, at 200% exposure, the advantage of the levered proportional strategy holds for risk aversion from $\gamma = 2$ to $\gamma = 10$ (Table C.VI), savings rates from 5% to 15% and withdrawal rates from 3% to 5% (Table C.VII), and, at 175%, in every quintile of the ex post real bill rate (Table C.VIII).⁹

6 Implementation

Individual investors can hold these strategies today; we verify it for a European household, using instruments available in September 2026. UCITS funds may reach 200% gross exposure (Committee of European Securities Regulators, 2010). The WisdomTree Global Efficient Core UCITS ETF holds 90% global equities and 60% government bond futures for 0.25% a year (WisdomTree, 2026), and the iMGP DBi Managed Futures UCITS ETF, listed in Paris since 2025, charges 0.75% (iM Global Partner, 2026). With physically backed gold, putting 74% of wealth in the first fund and 13% in each of the other two gives 66% stocks, 44% bonds, 13% gold, and 13% managed futures, a gross exposure of 137% financed inside the funds at futures-implied rates. The rest of the leverage, 29% of wealth for the proportional strategy at equal volatility and 32% for stocks and managed futures, can be borrowed through box spreads on index options, available in ordinary brokerage accounts and priced at 0.20% to 0.50% above SOFR or Treasury yields (CME Group, 2024; Wang, 2025). Our returns are before personal taxes.

⁹We take a partial equilibrium view. Large flows into gold would require price adjustments (Erb and Harvey, 2013), inelastic demand can make equity prices sensitive to flows (Gabaix and Koijen, 2023), and crowded trend-following trades can raise execution costs (Baltas, 2019).

7 Conclusion

We study the lifecycle investment problem of Anarkulova, Cederburg, and O’Doherty (2025) adding global bonds, gold, managed futures, and borrowing to the opportunity set. Diversification alone lowers the probability of ruin but not expected utility, and leverage alone adds little utility and raises ruin. Together, they improve all three measures. Levered to the volatility of unlevered equities, a multi-asset strategy matches the all-equity utility with a 6.35% savings rate instead of 10%, lowers the probability of ruin from 7.0% to 2.3%, and, at the same savings rate, funds the all-equity income more reliably. The gains come mainly from managed futures, which pay off in equity crises, rather than from leveraging bonds, and stocks with managed futures alone do better still. Equal utility is not equal income, preserving capital is not funding an income, and rankings differ at equal exposure and equal risk. With financing close to the bill rate, leveraging assets that pay off in equity crises is safer than concentrating in stocks.

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